WebRTC: The Way Forward for Video Over the Internet

Who doesn't love to make a free call? Web Real-Time Communication or WebRTC enables browser-to-browser communication for video calling, P2P file sharing and voice chat. This exciting, powerful and cutting-edge technology is set to revolutionise the way we communicate.

The goal of WebRTC is to provide real-time browser-to-browser voice and video communication without the need to download any plugins or software. So let's explore WebRTC architecture, the requirement for it in enterprises, and the leading market players who offer WebRTC as a service.

WebRTC is an open source project, released in May 2011 by Google, which allows browser based real-time communication. It supports browser-to-browser applications for video chat, voice calling and P2P file sharing without the need of any external or internal plugin. The WebRTC project is mainly supported by Google, Mozilla and Opera.

WebRTC components are accessed with JavaScript APIs—the Network Stream API, the PeerConnection API that allows two or more users to communicate browser-to-browser, and the DataChannel API that enables communication of other types of data for real-time gaming, text chat, file transfer, etc.

Connecting to remote peers uses NAT (network address translation) traversal technologies such as ICE (Interactive Connectivity Establishment), STUN (session traversal utilities for NAT), and TURN (traversal using relay NAT). STUN servers reside on the public Internet and have one simple task of checking the IP port address of an incoming request (from an application running behind a NAT) and sending that address back as a response. This process enables a WebRTC peer to get a publicly accessible address for itself, and then pass that on to another peer via a signalling mechanism, in order to set up a direct link. As many as 86 per cent of WebRTC calls successfully make a connection using STUN, though this can be less for calls between peers behind firewalls and complex NAT configurations.

RTCPeerConnection (PeerConnection API) of WebRTC tries to set up direct communication between peers over the User Datagram Protocol (UDP). If that fails, RTCPeerConnection resorts to Transmission Control Protocol (TCP). If that fails, TURN servers can be used